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RESEARCH INTERESTS

Privacy-preserving machine learning and trustworthy AI: differential privacy (central and local), federated learning under untrusted aggregation, and secure, manipulation-resistant LLM agents for privacy configuration, with applications to edge and Internet of Things (IoT) systems.

EDUCATION

University of Washington **Mar 2021 – Dec 2026 (expected)**
Ph.D. in Computer Science | GPA 3.98/4.0

- Dissertation: *UCLDP: A User-Centric Local Differential Privacy Model*
- Advisor: Dr. Eyhab Al-Masri, Associate Professor
- Advanced to Ph.D. candidacy: June 2024

University of Washington **Aug 2018 – Dec 2019**
M.S. in Computer Science and Systems | GPA 3.76/4.0

Xi'an Jiaotong University **Aug 2014 – Jun 2018**
B.Eng. in Automation | GPA 3.4/4.3

PUBLICATIONS

Google Scholar: 78 citations.

Journal Articles

[J1] **Yang, W.**, Al-Masri, E., & Kotevska, O. (2025). MIC-DP: A scalable correlation-aware differential privacy framework for high-dimensional data. *IEEE Transactions on Privacy, 2*, 131–143. https://doi.org/10.1109/TP.2025.3620376

[J2] Al-Masri, E., Souri, A., Mohamed, H., **Yang, W.**, Olmsted, J., & Kotevska, O. (2023). Energy-efficient cooperative resource allocation and task scheduling for Internet of Things environments. *Internet of Things, 23*, Article 100832. https://doi.org/10.1016/j.iot.2023.100832

Conference and Workshop Papers

[C1] Kotevska, O., **Yang, W.**, & Al-Masri, E. (2025). Engineering privacy at the edge: A practical guide to differential privacy in system architectures [Tutorial paper]. In *2025 IEEE 43rd International Conference on Computer Design (ICCD)* (pp. 872–875). IEEE. https://doi.org/10.1109/ICCD65941.2025.00128

[C2] **Yang, W.**, & Al-Masri, E. (2025, April 27). *Democratizing differential privacy: A participatory AI framework for public decision-making* [Workshop paper]. ACM CHI 2025 Workshop on Emerging Practices in Participatory AI Design in Public Sector Innovation, Yokohama, Japan. https://arxiv.org/abs/2504.21297

[C3] **Yang, W.**, & Al-Masri, E. (2024). ULDP: A user-centric local differential privacy optimization method. In *2024 IEEE World AI IoT Congress (AIIoT)* (pp. 316–322). IEEE. https://doi.org/10.1109/AIIoT61789.2024.10579023

[C4] Han, X., Wang, Z., & **Yang, W.** (2017). High-definition video image mosaics in real-time based on adaptive H-SIFT. In *Proceedings of the 2017 6th International Conference on Measurement, Instrumentation and Automation (ICMIA 2017)* (pp. 744–747). Atlantis Press. https://doi.org/10.2991/icmia-17.2017.134

Under Review

[U1] **Yang, W.**, Al-Masri, E., & Kotevska, O. (2026). *LDP-MIC: Correlation-aware local differential privacy for federated learning under untrusted aggregation* [Manuscript under review at *IEEE Transactions on Information Forensics and Security*].

In Preparation

[M1] **Yang, W.**, Al-Masri, E., & Wang, T. (2026). *When rules meet user preferences: Evaluating and verifying LLM agents for differential privacy configuration* [Manuscript in preparation; target venue: IEEE Symposium on Security and Privacy (S&P) 2027].

Posters

[P1] **Yang, W.**, Al-Masri, E., & Kotevska, O. (2026, August 12–14). *Secure privacy-budget configuration for LLM-powered edge AI systems* [Poster presentation]. 35th USENIX Security Symposium, Baltimore, MD, United States.

[P2] Al-Masri, E., & **Yang, W.** (2026, August 12–14). *ShadowBench-MCP: Measuring shadow invocations in MCP-style tool routing* [Poster presentation]. 35th USENIX Security Symposium, Baltimore, MD, United States.

[P3] **Yang, W.**, Al-Masri, E., & Kotevska, O. (2025, August 13–15). *Differential privacy meets data correlation: The MIC-DP framework* [Poster presentation]. 34th USENIX Security Symposium, Seattle, WA, United States.

RESEARCH EXPERIENCE

Doctoral Research, University of Washington **Sep 2022 – Present**
Advisor: Dr. Eyhab Al-Masri | In collaboration with Oak Ridge National Laboratory (Dr. Olivera Kotevska)

LLM Agents for Secure Differential Privacy Configuration *Manuscript targeting IEEE S&P 2027 [M1]*
With Dr. Tianhao Wang (University of Virginia) and Dr. Eyhab Al-Masri

- Designed a two-layer verifier (deterministic code checks plus an LLM checklist isolated from user input) that cut policy-violating privacy configurations on adversarial-pressure episodes from 4.6% to 0% across 9 frontier LLMs.
- Built DPConfig-Bench, a red-teaming benchmark for LLM agents configuring differential privacy: 52 scenarios, including 12 adversarial-pressure scenarios spanning 6 manipulation tactics (authority, urgency, policy doubt, precedent, emotional, reframing). Correct answers (minimal ε) are computed analytically and validated by Monte Carlo simulation.
- Deploying the agent as a ChatGPT plugin that gives users verified privacy-budget (ε) recommendations; public release in progress. First-author poster at USENIX Security 2026 [P1].

LDP-MIC: Federated Learning without a Trusted Aggregator *Under review, IEEE TIFS [U1]*

- Matched DP-SGD-level protection against gradient leakage attacks while keeping 5–8% higher accuracy: clients apply correlation-adaptive local differential privacy before sharing federated-learning updates, cutting reconstruction SSIM from 0.92 to 0.16 (an 83% reduction), with Opacus experiments on ORNL's Frontier exascale supercomputer.
- Engineered a batched federated-learning simulator in PyTorch (hand-written batched backpropagation across methods and clients) that executed 15.5K FL training runs (1.55M federated rounds) in about 4.5 hours on a 2-core CPU; validated against a NumPy reference with 37 unit tests, with hypotheses pre-registered in Git and Holm-corrected paired tests.

MIC-DP: Correlation-Aware Differential Privacy *IEEE Transactions on Privacy 2025 [J1]*

- Allocated per-feature privacy budget by maximal information coefficient (MIC) scores, outperforming uniform-noise differential privacy: ROC-AUC on Adult rose from 0.799 to 0.820 at $\varepsilon = 5$ (Holm-corrected $p < 0.001$) and from 0.801 to 0.823 at $\varepsilon = 10$ in a 30-trial follow-up evaluation.
- Reduced the noise needed to privately release correlation scores 650-fold by deriving a bounded-sensitivity fixed-grid MIC estimator (follow-up work).

- Released the method as an open-source, pip-installable Python package (`pip install mic-dp`); presented the framework as a poster at USENIX Security 2025 [P3].

Participatory AI and Conversational Agents for Privacy Decisions *ACM CHI 2025 Workshop [C2]*

- Proposed a participatory, human-centered AI framework that lets citizens shape differential privacy settings for public-sector AI through conversation.
- Designed PrivAgent (dissertation chapter), a conversational LLM agent grounded in bounded rationality that elicits users' privacy preferences.
- Designed and deployed an IRB-exempt online user study (UW IRB STUDY00025867; $N = 200$ participants on Prolific), with a web survey built in HTML/CSS/JavaScript and a Google Apps Script back end; results are reported in the dissertation.

ULDP: User-Centric Privacy-Budget Optimization *IEEE AIIoT 2024 [C3]*

- Automated privacy-budget (ε) selection for local differential privacy in edge computing with a TOPSIS-based multi-criteria optimization method that balances privacy, accuracy, and compute cost.

Research Collaboration with Microsoft (Dr. Minjia Zhang) **Sep 2021 – Jun 2022**
Ph.D. Research Project, University of Washington | *MigrationRouter: RL-Based Service Migration for Edge Computing*

- Built MigrationRouter, a reinforcement learning (Q-learning) agent that routes compute tasks across edge base stations, with an MDP formulation and a reward that trades off computational cost (CPU cycles) against migration cost (distance).
- Trained on Microsoft's Philly GPU-cluster traces: cleaned 117K DNN-training jobs to 102,666, joined them with real base-station locations, and processed the data in Google BigQuery.
- Reached 88% route accuracy (91% best-of-5) on 100-node networks and 99% route availability at 250+ nodes after hyperparameter tuning; built a web interface that maps the routes.
- Met weekly with Dr. Minjia Zhang (Microsoft) to review experiments and iterate on results, and co-built the dataset with a fellow researcher.

TEACHING EXPERIENCE

Graduate Teaching Assistant, University of Washington **2025**
School of Engineering and Technology

TCSS 573: Internet of Things	Winter 2025
TCSS 360: Software Development and Quality Assurance Techniques	Spring 2025
TCSS 445: Database Systems Design	Autumn 2025

- **Guest lecture (TCSS 559, Mar 2025):** research talk on MIC-based correlated differential privacy for students in Service Computing.
- **Guest lecture (TCSS 360, May 2025):** taught a 1.5-hour session on cloud computing, microservices, IaaS/PaaS/SaaS, Docker and containers, and Kubernetes to 40+ students, and designed a hands-on Docker in-class activity.
- Built hands-on tutorials for students in TCSS 360, TCSS 559, and TCSS 573.
- Ran group-project working sessions and code reviews, helped teams resolve Python environment issues, and gave feedback on project demos.

Grader, University of Washington **2019 – 2024**
School of Engineering and Technology | Graded assignments, held office hours, and organized and hosted group meetings for course projects

TCSS 573: Internet of Things	Winter 2022, 2023, 2024
TCSS 559: Service Computing	Autumn 2021, 2022, 2024
TCSS 445: Database Systems Design	Spring 2024
TCSS 460: Client/Server Programming for Internet Applications	Summer 2019, 2021, 2022

MENTORING AND LEADERSHIP

President, Peer Mentorship Association, Xi'an Jiaotong University (department level): mentored 100+ students over two years. 2015 – 2017

Team lead (team of 4), StreetSmarter IoT smart parking system: owned the system architecture, split work across hardware, back-end, and cloud tracks, and led the final pitch; won First Prize at the University of Washington Hackathon. 2019

PRESENTATIONS

Posters: two posters [P1], [P2] at the 35th USENIX Security Symposium, Baltimore, MD. Aug 2026

Research Showcase presentations, University of Washington. May 2026; Mar 2025; Mar 2024

Tutorial (co-presenter): Engineering privacy at the edge: A practical guide to differential privacy in system architectures [C1]. IEEE ICCD 2025, Richardson, TX. Nov 2025

Poster: Differential privacy meets data correlation: The MIC-DP framework [P3]. 34th USENIX Security Symposium, Seattle, WA. Aug 2025

Research talk: Maximum information correlated differential privacy: Enhancing utility in privacy-preserving data analysis. Guest lecture, TCSS 559 Service Computing. Mar 2025

Paper presentation: ULDP: A user-centric local differential privacy optimization method [C3]. 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA. May 2024

HONORS AND AWARDS

Outstanding Graduate Research Award, UW School of Engineering and Technology 2025 – 2026
Carwein-Andrews Award, University of Washington 2024
Graduates Merit Scholarship, University of Washington 2019, 2022, 2023
First Prize, University of Washington Hackathon (judged by the Microsoft Azure team) 2019
Undergraduate Leader Award, Xi'an Jiaotong University 2015
Excellent Undergraduate Award, Xi'an Jiaotong University 2014
Undergraduate Scholarship, Xi'an Jiaotong University 2014, 2016

PROFESSIONAL SERVICE AND MEMBERSHIPS

Reviewer, *Internet of Things and Cloud Computing* (journal) 2025 – 2026
Student Member, IEEE

SOFTWARE

mic-dp: open-source Python package for MIC-guided differential privacy (pypi.org/project/mic-dp) 2025

PROFESSIONAL EXPERIENCE

Neusoft Corporation **Jun 2018 – Aug 2018**
Software Engineer Intern | Shenyang, China

- Built the data layer of a government procurement platform at a leading Chinese IT solutions provider: designed the schema and storage structure, cleaned data, and contributed to front-end development.

TECHNICAL SKILLS

- **Privacy and security:** differential privacy (central/local, DP-SGD), federated learning, privacy attacks and defenses (gradient leakage), LLM red-teaming and guardrails, Opacus, Monte Carlo privacy verification
- **Machine learning:** PyTorch, reinforcement learning (Q-learning), LLM agents and evaluation, LLM APIs
- **Engineering:** Python, NumPy, pandas, matplotlib, Jupyter, unit testing, Git, statistical testing, SQL/SQLite, Google BigQuery, Flask, HPC (ORNL Frontier, AMD ROCm), NVIDIA CUDA, LaTeX